

24CS302- OBJECT ORIENTED PROGRAMMING USING JAVA

PART A QUESTON BANK

UNIT III

1) Define an abstract class in Java. Give an example.

An abstract class in Java is a class declared with the keyword `abstract`. It can have both abstract methods (without implementation) and concrete methods (with implementation). An abstract class cannot be instantiated directly; it must be inherited by a subclass that provides implementations for its abstract methods.

2) Explain the concept of multiple inheritance through interfaces in Java.

Java achieves multiple inheritance through interfaces. A class can implement multiple interfaces, thereby inheriting the abstract methods from all of them.

- Interfaces can be used to define multiple sets of behaviors.
- A single class can implement more than one interface using the **`implements`** keyword. □ The class must override all abstract methods of the interfaces.
- This allows Java to support multiple inheritance safely without ambiguity.

3)List some built in packages available in java and states its purpose.

`java.lang` - Contains fundamental classes such as `String`, `Math`, `System`, and `Object`. It is automatically imported in every Java program. |

`java.util` - Contains utility classes for data structures like `ArrayList`, `HashMap`, `Date`, and classes for collections and time operations. |

`java.io` - Provides classes for input and output operations, such as reading and writing files (`FileReader`, `BufferedWriter`, etc.). |

`java.net` - Supports networking functionalities such as creating sockets, URLs, and handling network

connections. |

`java.sql` - Contains classes and interfaces for database access using JDBC (e.g., **`Connection`**, **`Statement`**, **`ResultSet`**). |

`java.awt` - Used for creating ****Graphical User Interface (GUI)**** components like windows, buttons, and menus. |

`javax.swing` - Provides lightweight GUI components like **`JButton`**, **`JFrame`**, and **`JLabel`** for building modern desktop applications.

|

4) What are the different Access controls that can be applied to classes and methods in a package.

The main access modifiers are:

- ☐ **public`** – accessible from anywhere.
- ☐ **protected`** – accessible within the same package and subclasses.
- ☐ **(default)** – accessible within the same package.
- ☐ **private`** – accessible only within the same class.

5) What is the use of the `super` keyword in Java?

The **`super`** keyword in Java is used to refer to the immediate parent class object. It can be used to access parent class variables, methods, and constructors.

6) What is constructor chaining in Java?

Constructor chaining is the process of calling one constructor from another within the same class or from a parent class using **`this()`** or **`super()`**.

7) What is an abstract method?

An abstract method is a method without a body, declared using the **`abstract`** keyword. It must be implemented by subclasses of the abstract class.

8). What is an interface in Java?

An interface is a collection of abstract methods and constants that defines a contract for classes to implement. It supports multiple inheritance in Java.

9) How does Java achieve multiple inheritance using interfaces?

Java allows a class to implement multiple interfaces, thus achieving multiple inheritance, while avoiding the ambiguity issues found in multiple class inheritance.

10. What is a package in Java?

A package is a namespace that organizes related classes and interfaces into a single unit. It helps avoid naming conflicts and provides access control

11. Give one real-world example of a Java application using these concepts.

In an Employee Management System, abstract classes can define common employee behavior, interfaces can define payable or promotable roles, and `super` can call parent constructors to initialize base employee details.

UNIT IV

1) Differentiate between ArrayList and LinkedList in Java.

Feature	ArrayList	LinkedList
Data Structure	Uses a dynamic array to store elements.	Uses a doubly linked list to store elements.
Memory Usage	Requires less memory since it stores only data.	Requires more memory as it stores data along with node links (previous and next).
Access Time	Fast ($O(1)$) for accessing elements using index.	Slow ($O(n)$) because it must traverse nodes sequentially.
Insertion / Deletion	Slow for inserting or deleting in the middle (shifting elements needed).	Fast for insertion or deletion at any position (just change node links).

2) What is a user-defined exception? Provide a simple example.

A user-defined exception (or custom exception) in Java is an exception created by the programmer to handle specific application conditions that are not covered by built-in exceptions.

It is defined by extending the `Exception` class (for checked exceptions) or the `RuntimeException` class (for unchecked exceptions).

3)List any two classes used for reading and writing text files in Java

Two commonly used classes for reading and writing text files in Java are: FileReader -

Used to read characters from a text file.

FileWriter - Used to write characters to a text file.

4)Explain the use of Serializable interface in Java

The `Serializable` interface in Java is used to enable object serialization, which means converting an object into a byte stream so it can be saved to a file, sent over a network, or stored in a database.

Later, this byte stream can be deserialized (converted back into an object).

- The `Serializable` interface is part of the java.io package.
- The process uses ObjectOutputStream for writing and ObjectInputStream for reading objects.

5)List out the java Collections available to implement data structures in java applications

- **List** - `ArrayList`, `LinkedList`, `Vector`, `Stack`
 - Used to store ordered collections that allow duplicate elements.
- **Set** - `HashSet`, `LinkedHashSet`, `TreeSet`
 - Used to store unique elements (no duplicates).
- **Queue** - `PriorityQueue`, `ArrayDeque`, `LinkedList` Used to store elements in a FIFO (First-In-First-Out) order.
- **Deque** - `ArrayDeque`, `LinkedList`
 - Double-ended queue allowing insertion and removal at both ends.
- **Map** - `HashMap`, `LinkedHashMap`, `TreeMap`, `Hashtable` - Stores key–value pairs; keys are unique.

6)Write a java program by applying “for each loop” to iterate through elements.

```
public class ForEachExample {  
    public static void main(String[] args) {  
        // Array of integers
```

```

int[] numbers = {10, 20, 30, 40, 50};
// For-each loop to iterate through the array
System.out.println("Numbers in the array:");
for (int num : numbers) {
System.out.println(num);
}
}
}
}

```

7)What is mean File I/O what are the classes available in java to perform File input output operations

File I/O (Input/Output) in Java refers to the process of reading data from files (input) and writing data to files (output). It allows Java programs to persist data to files or read existing data for processing. File I/O in Java allows programs to persist and retrieve data using classes like `FileReader`, `FileWriter`, `BufferedReader`, `FileInputStream`, and `FileOutputStream`.

8)Write a syntax and example to create ArrayList in java.

Syntax

```
ArrayList<Type> listName = new ArrayList<>();
```

Type – Type of the elements stored(eg integer,String) listName – Name of the ArrayList

9. What is a user-defined exception in Java?

A user-defined exception in Java is a custom exception created by the programmer by extending the `Exception` class. It allows handling specific application-related errors that are not covered by built-in exceptions, providing more meaningful error messages and control over program flow.

10. What is nested try-catch in Java?

Nested try-catch refers to placing a try-catch block inside another try or catch block. It allows handling exceptions at multiple levels, providing specific handling for different operations while maintaining program stability.

11. What are the main interfaces in the Java Collections Framework? The main interfaces in Java Collections Framework are `List`, `Set`, and `Map`. `List` is used for ordered collections with duplicates, `Set` stores unique elements, and `Map` stores key-value pairs with unique keys.

12. What is an ArrayList in Java?

`ArrayList` is a resizable array implementation of the `List` interface that allows duplicate elements and maintains insertion order. It provides dynamic memory allocation, fast random access using indices, and is part of the `java.util` package.

13. What is a LinkedList in Java?

`LinkedList` implements the `List` and `Deque` interfaces and stores elements as nodes in a doubly linked list. It allows fast insertion and deletion at any position, supports iteration, and is useful when frequent modifications are required.

14. What is a HashSet in Java?

`HashSet` implements the `Set` interface and stores unique elements without maintaining any specific order. It provides efficient operations for adding, removing, and searching elements based on hashing.

15. What is a TreeSet in Java?

`TreeSet` implements the `Set` interface and stores unique elements in **sorted order** according to natural ordering or a specified comparator. It is useful when automatically maintaining sorted collections is required.

16. What is a HashMap in Java?

`HashMap` implements the `Map` interface and stores data as key-value pairs where keys are unique and values can be duplicated. It provides fast access and modification using hashing and is widely used for

mapping relationships.

17. What is an Iterator in Java?

An `Iterator` is an object that allows sequential traversal of a collection. It provides methods like `hasNext()` and `next()` to iterate safely and remove elements during iteration without causing `ConcurrentModificationException`.

18. What is the enhanced for loop in Java?

The enhanced for loop, also called the for-each loop, is used to iterate through arrays or collections without using an index. It improves code readability and reduces errors by automatically traversing all elements sequentially.

19. Which classes are used for reading and writing text files in Java?

Text file I/O in Java is commonly performed using `FileReader` and `BufferedReader` for reading, and `FileWriter` and `BufferedWriter` for writing. Buffered classes improve efficiency by reducing the number of I/O operations.

20. What is Serialization in Java?

Serialization in Java is the process of converting an object into a byte stream so it can be saved to a file or sent over a network. Deserialization restores the object from the byte stream, and classes must implement the `Serializable` interface to enable this feature.

UNIT V

1) List the states of a thread in Java.

In Java, a thread can exist in six states as defined by the `Thread.State` enum: New, Runnable, Running, Blocked, Waiting, and Terminated.

A thread is in the New state when it is created but not yet started. Once the `start()` method is called, it moves to the Runnable state, waiting for the thread scheduler to allocate CPU time. When the thread is executing, it is in the Running state. If the thread is waiting to acquire a lock, it enters the Blocked state, and if it is waiting indefinitely for another thread to signal it, it is in the Waiting state. Finally, when the thread completes execution, it reaches the Terminated state, after which it cannot be restarted.

2) What is an event delegation model in Java GUI programming.

In Java GUI programming, the event delegation model is a design pattern used to handle user interactions such as button clicks, mouse movements, or key presses. Instead of each GUI component handling its own events, the component generates an event object and delegates it to a separate listener object that implements a corresponding event listener interface (like `EventListener` or `MouseListener`).

3) Mention any two layout managers in Java Swing.

Two commonly used layout managers in Java Swing are:

1. `FlowLayout` – Arranges components in a left-to-right flow, much like words in a paragraph.

Components wrap to the next line when the current line is full.

2. `BorderLayout` – Divides the container into five regions: North, South, East, West, and Center.

Components are added to these specific regions, and each region can contain only one component.4)

What is the purpose of ActionListener in Java?

In Java, the `ActionListener` interface is used to handle action events generated by GUI components such as buttons, menu items, or text fields. When a user performs an action like clicking a button, the component notifies the registered `ActionListener` by calling its `actionPerformed()` method. This allows developers to define specific responses to user actions, enabling interactive and event-driven behavior in GUI applications.

5) What is the need of Inter thread communication in java In Java, inter-thread communication is needed when multiple threads share a common resource and must coordinate their execution to avoid conflicts or data inconsistency. For example, in a producer-consumer scenario, the producer thread generates data while the consumer thread uses it. Inter-thread communication using methods like `wait()`, `notify()`, and `notifyAll()` allows threads to safely exchange information,

6) State the classes used to handle GUI events in java

In Java, GUI events are handled using listener interfaces and event classes provided in the `java.awt.event` package. Common listener interfaces include

□ `ActionListener` (for button clicks),

- 'MouseListener' and 'MouseMotionListener' (for mouse events),
- 'KeyListener' (for keyboard events), and
- 'WindowListener' (for window events).

The corresponding event classes such as 'ActionEvent', 'MouseEvent', 'KeyEvent', and 'WindowEvent' encapsulate information about the event, which is passed to the listener methods to respond appropriately in the application.

7) Write the purpose of MouseListener Interface in java

The 'MouseListener' interface in Java is used to handle mouse events on GUI components, such as clicking, pressing, releasing, entering, or exiting a component. By implementing this interface, a class can define custom responses to these mouse actions through its methods ('mouseClicked()', 'mousePressed()', 'mouseReleased()', 'mouseEntered()', and 'mouseExited()'), enabling interactive behavior and enhancing user interaction in graphical applications.

8) Mention any two Swing components used to build forms.

Two commonly used Swing components for building forms in Java are: 1. JTextField – A single-line text field that allows the user to enter or edit text. 2. JButton – A clickable button that can trigger actions such as submitting a form or performing a task when clicked.

9. What is AWT in Java?

* AWT (Abstract Window Toolkit) is Java's original GUI toolkit for building window-based applications.* It provides components like buttons, labels, and text fields to create user interfaces.

* AWT components are platform-dependent, meaning their appearance depends on the underlying operating system.

10. What is Swing in Java?

* Swing is a lightweight GUI toolkit that extends AWT for building rich user interfaces.

* It provides components like 'JButton', 'JLabel', 'JTextField', and 'JTable'. * Swing

components are platform-independent and support a pluggable look and feel. **11. What are**

Layout Managers in Java?

* Layout Managers control the arrangement of components within a container. *

Examples include `FlowLayout`, `BorderLayout`, and `GridLayout`.

* They help create responsive and organized GUIs without manually setting component positions. **12. What**

is the Event Delegation Model in Java?

* Event Delegation Model separates event sources from event handlers. * Components

generate event objects which are passed to listener objects. * It improves modularity

and makes handling multiple events more efficient. **13. What is the purpose of**

ActionListener in Java?

* `ActionListener` handles action events like button clicks or menu selections. * The

`actionPerformed()` method is called whenever the event occurs. * It enables interactive

behavior in GUI applications.

14. Name two common Swing components used in forms.

* `JTextField` – allows users to enter or edit single-line text.

* `JButton` – a clickable button that triggers actions when pressed.

* Other examples include `JLabel` (display text) and `JPasswordField` (secure text input).

15. Give examples of applications built using Java GUI.

Chat applications – allow real-time messaging between users.

Form-based systems – such as registration forms or contact entry forms.

Simulation apps – visualizing processes or models with interactive interfaces.